

Designing Markets

Chapter 4: Handbook of Market Design

Summary Presentation

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Overview: Three Goals of Market Design

- 1 **Diagnose** potential market failures and identify their causes
- 2 **Evaluate** markets and compare alternative designs (qualitative and quantitative effects)
- 3 **Propose** new, improved designs building on lessons learned

Key insight: Seemingly small details about market rules can have large impacts on market performance

4.1 Diagnosing Market Failures

Common types of failures documented:

- **Market Power:** Exercise by large participants (US Spectrum: 7-20% higher payouts)
- **Collusion:** Centralized mechanisms viewed with suspicion (medical residency lawsuit)
- **Rent Seeking:** Wasteful competition for rents (HFT, ticket brokers)
- **Participation/Entry:** Adding one bidder $>$ optimal design (Bulow-Klemperer)
- **Fragmentation:** Kidney exchange fragmentation reduces transplants 30%+
- **Search Costs:** Students applying to too few/many schools
- **Thin markets:** Low participation $\rightarrow$ hard to match
- **Congestion:** Too many participants/transactions $\rightarrow$ delays/costs
- **Information frictions:** Hidden quality, hidden preferences
- **Market Power:** Strategic Behavior, hidden preferences
- **Strategic Misreporting:** Manipulable mechanisms $\rightarrow$ inefficiency

4.2 Evaluating and Comparing Designs

Core Challenge: Theory often ambiguous—need empirical quantification of trade-offs

School Choice (NYC Reform):

- Centralization achieved 80% of potential gains (old: $\leq 33\%$)
- **Key insight:** Coordination matters more than specific mechanism choice
- DA vs. IA: Similar welfare if equilibrium play, but mistakes favor DA

Dynamic Assignment (organs, housing, licenses):

- Challenge: Objects arrive over time, can't elicit full preferences
- FCFS vs. LCFS depends on preference structure (idiosyncratic vs. vertical)
- Approach: Dynamic discrete choice models + administrative data

4.2 Methodological Approaches

Beyond Structural Models:

- **Direct comparison:** Two designs in same market (Niederle-Roth: gastro fellowship)
- **Survey-based:** Elicit truthful preferences, simulate (Budish-Cantillon)
- **Descriptive:** Document post-reform changes (Prendergast: food banks)

Non-Utilitarian Objectives:

- Public housing: targeting vs. efficiency; Organs: survival vs. urgency
- Distributional concerns and meritocratic admissions
- **Challenge:** Selection bias when choices correlate with outcomes
- **Solution:** Combine choice models with treatment effect estimation

Note: Centralized markets provide ideal data for policy evaluation

4.3 Proposing New Designs: Medical Match & School Choice

Medical Match (NRMP):

- Matches 25,000 grads annually; discovered Gale-Shapley in 1950s
- Solved market unraveling; Roth-Peranson: handled married couples

School Choice (Abdulkadiroğlu-Sönmez 2003):

- Documented Boston mechanism's manipulability
- Proposed DA and TTC as alternatives
- DA widely adopted; efficiency differences small but complexity/fairness matter

4.3 Proposing New Designs: Course Allocation & Other Markets

A-CEEI (Budish):

- Allocate oversubscribed courses efficiently and fairly
- Approximately efficient, strategy-proof; satisfies envy-freeness
- Multi-paper program: theory, computation, experiments, implementation

Food Banks (Prendergast):

- Pseudomarket with internal currency → equitable, efficient sorting

Financial Exchanges (Budish et al.):

- Frequent batch auctions eliminate speed arbitrage
- Could reduce costs 17% (\$5B/year); trade on price not speed